

《计算机网络》作业

第九次作业：（英文教材第五章 20, 21, 34, 35, 补充题 1）

20. (5.6.1 IPv4 Packet Format) Suppose that host A is connected to a router R1, R1 is connected to another router, R2, and R2 is connected to host B. Suppose that a TCP message that contains 900 bytes of data and 20 bytes of TCP header is passed to the IP code at host A for delivery to B. Show the Total length, Identification, DF, MF, and Fragment offset fields of the IP header in each packet transmitted over the three links. Assume that link A-R1 can support a maximum frame size of 1024 bytes including a 14-byte frame header, link R1-R2 can support a maximum frame size of 512 bytes, including an 8-byte frame header, and link R2-B can support a maximum frame size of 512 bytes including a 12-byte frame header.
21. (5.6.1 IPv4 Packet Format) A router is blasting out IP packets whose total length (data plus header) is 1024 bytes. Assuming that packets live for 10 sec, what is the maximum line speed the router can operate at without danger of cycling through the IP datagram ID number space?
34. (5.6.1 IPv4 Packet Format) Most IP datagram reassembly algorithms have a timer to avoid having a lost fragment tie up reassembly buffers forever. Suppose that a datagram is fragmented into four fragments. The first three fragments arrive, but the last one is delayed. Eventually, the timer goes off and the three fragments in the receiver's memory are discarded. A little later, the last fragment stumbles in. What should be done with it?
35. (5.6.1 IPv4 Packet Format) In IP, the checksum covers only the header and not the data. Why do you suppose this design was chosen?

补充题 1(5.6.1 IPv4 Packet Format) 某主机 HostA 的以太网卡收到两个 IP 包，IP 包的前 20 个字节如下图所示。

45	00	05	dc	76	ea	02	00	0a	11	82	df	b8	9e	7e	a1	c3	a8	02	1f
45	00	03	af	76	ea	01	72	0a	11	a3	9a	b8	9e	7e	a1	c3	a8	02	1f

图 IPv4 包

请回答如下问题

- (1) 主机 HostA 的 IP 地址是多少（用十进制点分制表示）
- (2) 第一个 IP 包的上层协议是什么？首部长度是多少字节？
- (3) IPv4 数据包中哪几个域与分片相关？如何判断 IP 包是否被分片了？
- (4) 图中所示两个 IP 包是否属于同一个原始 IP 包，为什么？是否可以计算原始 IP 包的数据长度？如果可以，请计算长度，如果不可以，请说明理由。
- (5) 根据这两个 IP 包主机 HostA 是否可以重组原始 IP 包数据并提交给上层？请说明理由。